



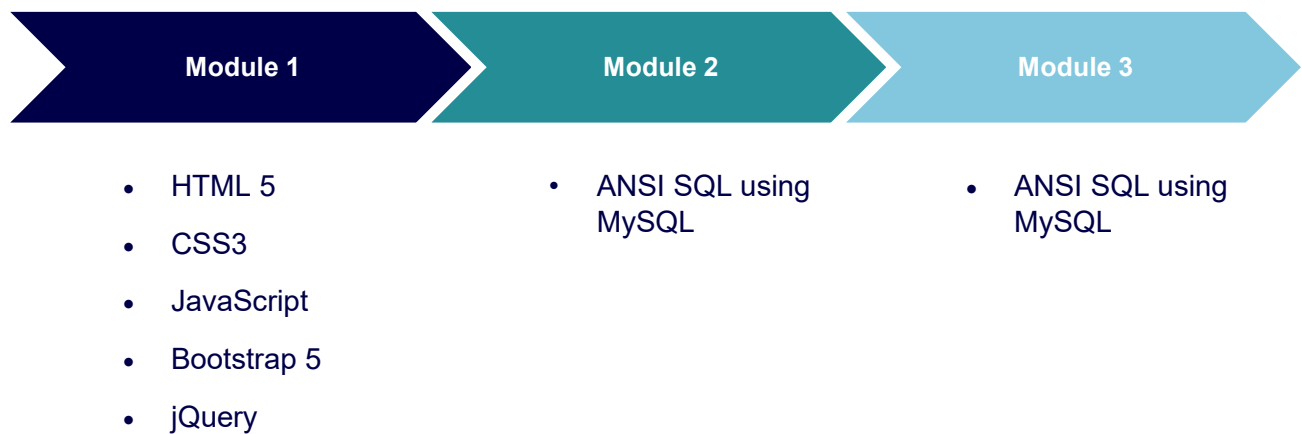
Digital Nurture 5.0

DotNet FSE - Upskilling Handbook

Program Highlights

- The Upskilling learning program has 3 Modules to be completed. Go through the recommended self- learning resources and practice the exercises to excel in the recommended DotNet FSE modules.
- Upskilling covers the below mentioned skills.
 - **HTML 5 | CSS3 | JavaScript | Bootstrap 5 | jQuery | ANSI SQL using MySQL | C# | ADO.NET**
- Please aim to complete the learning in 3 weeks time.

Recommended Program Sequence



The learning journey contains the following modules, followed by an Upskilling Final Assessment

Learning Approach

Self-paced learning through open-source learning reference links



Self-paced learning using open-source reference links

- Please refer to the learning reference links provided to learn and understand the recommended concepts.
- We expect you to dedicate **2-3 hours of focused attention daily** and **schedule longer sessions on weekends** for your learning modules.
- Balance your learning with regular academic commitments.

***Disclaimer:** Cognizant does not claim ownership or responsibility for the content or any issues with the links provided, as they are merely references available on the internet. Candidates are free to leverage additional sources beyond what has been provided to enhance their skill capabilities for upskilling.*

Effective Learning Strategies

Create a Study Schedule	Set Clear Goals	Stay Organized	Active Learning
• Dedicate specific times daily for learning.	• Define what you want to achieve. • Break down tasks into manageable chunks.	• Keep track of your progress and deadlines. • Use tools like calendars, to-do lists, and reminders.	• Engage with the material through hands-on exercises. • Practice coding and solving problems regularly. • Ensure to complete all exercises.

Online Code Editor

Online Code Editors	URL
HTML5, CSS3, JavaScript	https://www.w3schools.com/tryit/tryit.asp?filename=tryhtml_hello
	https://onecompiler.com/html
	https://jsfiddle.net/
Bootstrap5	https://www.w3schools.com/bootstrap5/bootstrap_editor.php
jQuery	https://www.w3schools.com/jquery/jquery_editor.asp
SQL	https://www.mycompiler.io/new/sql
	https://www.jdoodle.com/execute-sql-online/
	https://www.tutorialspoint.com/execute_sql_online.php
	https://www.w3schools.com/sql/trysql.asp?filename=trysql_asc
	https://onecompiler.com/mysql
Dot Net	https://try.dot.net/
	https://dotnetfiddle.net/
	https://onecompiler.com/csharp
	https://ideone.com/
• Set up Integrated Development Environments (IDEs) like VS Code. • Install necessary tools and libraries as per course requirements.	

Exercise Instructions

In this learning program, you will be required to complete exercises designed to reinforce the concepts learned. These exercises are hosted on a public GitHub repository and must be downloaded and solved. Follow the instructions below to ensure a smooth and productive exercise workflow:

1. Access the Exercises:

- Exercises will be made available in our public GitHub repository.
- The repository URL is **<https://github.com/seshadrimr/Digital-Nurture-DotNetFSE>**
- Navigate to the repository and locate the folder named ***DotNet FSE -> Upskilling***. Inside it, you will find a set of exercises.

2. Download the Exercises:

- Download the files for the exercises by clicking the download button on the GitHub repository page.

3. Solving the Exercises:

- Solve the problem statements provided in the downloaded files.
- Ensure that you understand the problem requirements and apply the concepts learned.
- Take your time to think through the solutions and code them accurately.

4. Self-Evaluation:

- After completing the exercises, evaluate your solutions based on the problem criteria.
- Compare your approach with any hints or solutions provided if available.
- Reflect on any mistakes or areas where you can improve.

5. Submitting Solutions:

- Firstly, organize your solutions and keep them in a folder.
- Create a public repository in your personal GitHub account, upload your solution folder and share the URL with the POC on demand.

6. Additional Support:

- If you encounter difficulties or have questions about the exercises, seek help from peers.
- Utilize the resources and links provided in this handbook for further assistance.

Module 1

HTML 5

Overview:

This module provides a comprehensive foundation in HTML5, the core markup language for web development. Learners will explore the essential concepts, elements, and best practices of HTML5 while gaining practical hands-on experience using modern development tools such as Visual Studio Code and Google Chrome Developer Tools. The module also introduces critical HTML5 APIs like Web Storage, Geolocation, and client-side database handling, enabling learners to create dynamic, interactive, and user-friendly web applications.

Through real-world examples, exercises, and exploration of browser features, participants will not only learn to structure web content effectively but also enhance the user experience with interactivity and persistence.

Learning Objectives:

By the end of this module, learners will be able to:

- Understand and Set Up the HTML5 Environment
- Apply HTML5 Tags for Content Structuring
- Build Navigation and References in Web Pages
- Handle Events and Interactions
- Explore Advanced HTML5 Form Features
- Implement Web Storage Solutions
- Understand Client-Side Databases
- Utilize HTML5 Geolocation API

Self-Learning (Open-source links):

HTML 5		
Topics	Sub-Topics	Learning Reference Links
Introduction	Need and Benefits of HTML, Setup, Browser, BOM and DOM, DOCTYPE, Character Encoding, <script> , <link>, HTML5 Document, Comments	https://www.w3schools.com/html/html_intro.asp
Getting Started	Visual Studio Code features, Google Chrome Developer tools, Inspect document	https://learn.microsoft.com/en-us/microsoft-edge/visual-studio-code/microsoft-edge-devtools-extension/open-devtools-and-embedded-browser
Elements & Attributes	Formatting Tags, List, Table, Form & Input Tags, Images, Styles, placeholder, inline and block elements, id vs class attributes	https://www.w3schools.com/tags/
		https://www.geeksforgeeks.org/html-forms/
		https://www.geeksforgeeks.org/difference-between-id-and-class-attributes-in-html/
		https://www.w3schools.com/html/html_blocks.asp
Navigation		https://www.w3schools.com/tags/tag_nav.asp

	Navigation tags, hyper link, reference to intermediate section	https://www.htmldog.com/guides/html/intermediate/sectioning/
Events	onblur, onchange, onclick, form related events, load events, key events, mouse events, ondblclick, onbeforeunload, oncanplay	https://www.w3schools.com/jsref/dom_obj_event.asp
Web Forms 2.0	<input> element in HTML5, <output>, placeholder attribute, autofocus, required	https://www.tutorialspoint.com/html5/html5_web_forms2.htm
Web Storage	Session storage, local storage and Delete web storage	https://www.w3schools.com/html/html5_webstorage.asp
Web SQL Database	openDatabase, transaction, transaction	https://www.geeksforgeeks.org/what-is-web-sql/
Geo location	Geolocation Methods, Location Properties, Handling Errors, Position Options	https://www.geeksforgeeks.org/html-geolocation/

Check Your Understanding:

[HTML 5 Quiz](#)

CSS3

Overview:

This module introduces learners to the fundamental concepts of CSS3, a cornerstone technology for designing visually engaging, responsive, and user-friendly websites. Starting from understanding the need and benefits of CSS, learners will explore how to set up and apply CSS to HTML documents effectively. The module covers basic syntax, commenting, and the three primary ways of including CSS: inline, embedded, and external. Building on this foundation, the course dives into CSS3 selectors,

enabling learners to target and style HTML elements efficiently. It then covers essential properties for managing color, backgrounds, fonts, text formatting, hyperlinks, lists, and tables.

The module also introduces the critical concept of the CSS Box Model, explaining margins, padding, borders, outlines, and the difference between visibility and display properties. Learners will further explore layout enhancements like multiple columns.

Finally, the module culminates with an introduction to Media Queries and Responsive Web Design (RWD) — crucial for creating adaptable websites that look great on devices of all sizes.

Learning Objectives:

By the end of this module, learners will be able to:

- Explain the need for CSS and identify its key benefits in web development.
- Set up CSS correctly and apply CSS using inline styles, embedded styles, and external stylesheets.
- Understand and apply CSS syntax rules and use comments effectively to enhance code readability.
- Utilize different types of CSS selectors (universal, element type, ID, class, grouping) to target HTML elements.
- Style web page elements using properties related to color, backgrounds, fonts, text, links, lists, and tables.
- Explain and apply the CSS Box Model, including margin, padding, border, and outline properties.
- Differentiate between the visibility and display properties and use them appropriately.
- Implement multiple column layouts using CSS3 properties.
- Create responsive designs using Media Queries to adapt websites for different screen sizes and devices.
- Demonstrate fundamental Responsive Web Design (RWD) principles to ensure website accessibility and usability across devices.

Self-Learning (Open-source links):

CSS3		
Topics	Sub-Topics	Learning Reference Links
Introduction	Need and Benefits of CSS, Setup, CSS Syntax, CSS Comments, Including CSS in HTML Documents(Inline styles, Embedded styles, External style sheets)	https://www.w3schools.com/html/html_css.asp
		https://www.w3schools.com/css/css_howto.asp
		https://www.geeksforgeeks.org/types-of-css-cascading-style-sheet/
Selectors	CSS3 Selectors - Universal Selector, Element Type Selector, Id Selectors, Class Selectors, Grouping Selectors	https://www.geeksforgeeks.org/css-selectors/
Styling	CSS Color, CSS Background, CSS Fonts, CSS Text, CSS Links, CSS Lists, CSS Tables	https://www.w3schools.com/css/css_colors.asp
		https://www.w3schools.com/css/css_background.asp
		https://www.w3schools.com/css/css_font.asp
		https://www.w3schools.com/css/css_link.asp
		https://www.w3schools.com/css/css_list.asp
		https://www.w3schools.com/css/css_table.asp
Box Model	CSS Box Model, Margin, padding, border, Outline, Visibility vs. Display, Multiple Columns	https://w3schools.com/css/css_boxmodel.asp
		https://www.geeksforgeeks.org/what-is-the-difference-between-visibilityhidden-and-displaynone/
Advanced	Media Queries, RWD	https://www.w3schools.com/css/css3_mediaqueries.asp
		https://www.w3schools.com/css/css_rwd_intro.asp

Check Your Understanding:

[CSS3 Quiz](#)

JavaScript

Overview:

This module introduces learners to the core concepts and modern practices of JavaScript, one of the most essential languages for web development. Starting from setting up a development environment, learners will explore JavaScript syntax, data handling, and control structures. The module then progresses into functions, objects, and the Document Object Model (DOM) for dynamic web interactions.

Advanced topics such as asynchronous programming, modern ES6+ features, and working with forms and APIs are covered to equip learners with practical skills for real-world applications. The module concludes with an overview of debugging techniques, testing strategies, and an introduction to popular JavaScript frameworks and libraries, including using jQuery for DOM manipulation.

Learning Objectives:

By the end of this module, learners will be able to:

- Understand the role and setup of JavaScript in modern web development environments.
- Apply fundamental programming concepts using JavaScript syntax, data types, operators, and control structures.
- Develop modular and maintainable code through the use of functions, scope management, and closures.
- Create dynamic web applications by manipulating objects, handling events, and interacting with the DOM.
- Manage asynchronous operations effectively using callbacks, Promises, and async/await patterns.
- Leverage modern JavaScript features such as ES6+ syntax enhancements and modular programming techniques.
- Work with forms and external data sources by implementing form handling and API communication using Fetch API.
- Debug and test JavaScript applications to ensure code reliability and performance.
- Explore and utilize JavaScript libraries and frameworks to accelerate web development and improve productivity.

Self-Learning (Open-source links):

JavaScript		
Topics	Sub-Topics	Learning Reference Links
Introduction to JavaScript	What is JavaScript?, Setting Up JavaScript Development Environment	https://www.geeksforgeeks.org/introduction-to-javascript/?ref=lbp
JavaScript Basics	Syntax and Statements, Data Types, Operators	https://www.geeksforgeeks.org/javascript-basic-syntax/?ref=lbp
		https://www.geeksforgeeks.org/variables-datatypes-javascript/?ref=lbp
		https://www.geeksforgeeks.org/javascript-operators/?ref=lbp
Control Flow	Conditional Statements, Loops, Error Handling	https://www.geeksforgeeks.org/control-statements-in-javascript/
Functions and Scope	Function Basics, Scope and Closures, Higher-Order Functions	https://www.w3schools.com/js/js_functions.asp
		https://www.geeksforgeeks.org/difference-between-scope-and-closures-in-javascript/
		https://www.geeksforgeeks.org/javascript-higher-order-functions/
Objects and the Document Object Model (DOM)	Objects in JavaScript, JavaScript Prototypes, The DOM, Event Handling	https://www.geeksforgeeks.org/dom-document-object-model/
		https://www.w3schools.com/js/js_events.asp
Arrays and Array Methods	Array Basics, Array Methods	https://w3schools.com/js/js_arrays.asp
		https://www.w3schools.com/js/js_array_methods.asp
Asynchronous JavaScript	Understanding Asynchronous Programming, Callbacks and Callback Hell, Promises,	https://developer.mozilla.org/en-US/docs/Learn_web_development/Extensions/Async_JS/Introducing
		https://geeksforgeeks.org/javascript-promise/
		https://www.geeksforgeeks.org/what-to-understand-callback-and-callback-hell-in-javascript/

	Async/Await Syntax	
JavaScript ES6+ Features	Let and Const, Template Literals, Destructuring, Rest and Spread Operators, Modules, Default Parameters	https://www.w3schools.com/js/js_es6.asp
JavaScript in Web Development	Working with Forms, AJAX and Fetch API	https://www.infoworld.com/article/2169473/using-javascript-and-forms.html
		https://www.w3schools.com/jsref/api_fetch.asp
		https://www.w3schools.com/js/js_ajax_intro.asp
Debugging and Testing JavaScript	Debugging Tools, Testing JavaScript Code	https://www.w3schools.com/js/js_debugging.asp
		https://learn.microsoft.com/en-us/visualstudio/javascript/unit-testing-javascript-with-visual-studio?view=vs-2022&tabs=jest
Introduction to JavaScript Frameworks and Libraries	Overview of Popular JavaScript Frameworks, Using jQuery for DOM Manipulation	https://www.geeksforgeeks.org/most-popular-javascript-frameworks-for-web-development/
		https://www.w3schools.com/jquery/jquery_intro.asp
		https://www.tutorialspoint.com/jquery/jquery-dom.htm

Check Your Understanding:

[JavaScript Quiz](#)

Bootstrap 5

Overview:

This module provides a comprehensive introduction to Bootstrap 5, the world's most popular front-end framework for building responsive, mobile-first websites. You will start by understanding the core structure and setup of Bootstrap, followed by mastering the responsive grid system and flex utilities. The module will guide you through working with essential components like typography, forms, buttons, navbars, cards, and media objects. You'll also explore utility classes that make rapid UI development efficient, including spacing, colors, visibility, and positioning. Finally, the module covers using Bootstrap Icons, enhancing interactivity with JavaScript plugins, and customizing Bootstrap using Sass for project-specific styling needs.

Learning Objectives:

By the end of this module, learners will be able to:

- Set up a new project using Bootstrap 5 and understand its folder structure and core files.
- Use Bootstrap's Grid System to create flexible, responsive layouts.
- Apply column classes, alignment, and reordering techniques for different screen sizes.
- Utilize responsive flexbox utilities for efficient layout management.
- Style text effectively using Bootstrap Typography utilities.
- Build and customize forms, buttons, navbars, cards, and media objects.
- Manage spacing, colors, backgrounds, borders, shadows, and rounded corners using Bootstrap utility classes.
- Control element visibility, display properties, and positioning for responsive design.
- Integrate and use Bootstrap Icons in web applications.
- Utilize Bootstrap's JavaScript plugins to add dynamic behaviors such as modals, dropdowns, and carousels.
- Understand the basics of customizing Bootstrap using Sass variables and mixins to align with project-specific design requirements.

Self-Learning (Open-source links):

BootStrap 5		
Topics	Sub-Topics	Learning Reference Links
Introduction to Bootstrap 5	Overview of Bootstrap Framework, Setting Up Bootstrap 5, Bootstrap Structure and Files	https://www.w3schools.com/bootstrap5/bootstrap_get_started.php
Bootstrap Grid System	Fundamentals of Responsive Grid Layout, Column Layouts and Grid Classes, Alignment and Reordering in Grid, Responsive Flexbox Utilities	https://www.w3schools.com/bootstrap/bootstrap_grid_system.asp
Bootstrap Components	Typography, Forms, Buttons, Navbars and Navigation, Cards and Media Objects	https://www.geeksforgeeks.org/explain-the-components-of-bootstrap/
Bootstrap Utilities and Helpers	Spacing Utilities, Colors and Backgrounds, Display and Visibility, Borders, Shadows, and Rounded Corners, Positioning Utilities	https://www.w3schools.com/bootstrap5/bootstrap_utilities.php
Advanced Bootstrap 5 Features	Icons with Bootstrap Icons, Bootstrap 5 JavaScript Plugins, Customization with Sass	https://www.geeksforgeeks.org/how-to-add-icons-in-project-using-bootstrap/
		https://getbootstrap.com/docs/5.0/getting-started/javascript/
		https://www.freecodecamp.org/news/how-to-customize-bootstrap-with-sass/

Check Your Understanding:

[Bootstrap 5 Quiz](#)

Overview:

This module provides a foundational understanding of jQuery - a lightweight, fast, and feature-rich JavaScript library. Designed to simplify HTML document traversal, event handling, animation, and AJAX interactions, jQuery helps developers write less code while achieving more functionality. Learners will explore the core features of jQuery, understand how to include the library in web pages, and get hands-on with practical DOM manipulation and event handling techniques. This module aims to equip learners with the skills needed to make web pages interactive and dynamic using jQuery.

Learning Objectives:

By the end of this module, learners will be able to:

- Define what jQuery is and explain its purpose and benefits in web development.
- List the key features of jQuery, including simplicity, cross-browser support, and chainable methods.
- Include the jQuery library in a project using both local and CDN methods.
- Use the `$()` function to select and interact with HTML elements efficiently.
- Apply various jQuery methods to manipulate content, attributes, CSS, and classes.
- Perform DOM manipulation to dynamically update or modify web page elements.
- Handle user interactions using built-in event methods such as `click()`, `hover()`, `submit()`, and others.
- Utilize event helper methods for concise and effective event handling.

Self-Learning (Open-source links):

jQuery		
Topics	Sub-Topics	Learning Reference Links
Introduction to jQuery and its Features	What is jQuery? Features of jQuery	https://www.w3schools.com/jquery/jquery_intro.asp
Basic Components of jQuery	jQuery Library Inclusion, The <code>\$()</code> Function, jQuery Methods	https://www.w3schools.com/Jquery/jquery_events.asp
DOM Manipulation and Events in jQuery	DOM Manipulation, Working with Events, Event Helpers	https://www.tutorialspoint.com/jquery/jquery-dom.htm

Check Your Understanding:

[jQuery Quiz](#)

Module 2

ANSI SQL using MySQL

Overview:

This module provides a comprehensive introduction to ANSI SQL standards with practical application using MySQL. It covers the importance of SQL standardization, basic to advanced querying techniques, data manipulation operations, and schema management. Learners will start by understanding the foundation of SQL and gradually move into querying data with SELECT statements, applying filters, sorting results, using aggregate functions, and performing table joins. The module further explores working with subqueries, manipulating data within tables, and managing database schema objects. It also focuses on ensuring performance and integrity through indexes and constraints. Upon completion, learners will have a strong foundational knowledge to work with relational databases using SQL confidently and efficiently.

Learning Objectives:

By the end of this module, learners will be able to:

- Understand the role and significance of ANSI SQL and MySQL in relational database management.
- Retrieve and manipulate data from relational databases using standard SQL queries.
- Apply filtering, sorting, and aggregation techniques to manage query outputs effectively.
- Implement advanced querying techniques using joins and subqueries to combine and analyze complex data sets.
- Perform data manipulation operations including inserting, updating, and deleting records.
- Design and manage database structures by creating, altering, and dropping tables.
- Enhance database performance and enforce data integrity using indexes and constraints.

Self-Learning (Open-source links):

ANSI SQL using MySQL		
Topics	Sub-Topics	Learning Reference Links
Introduction to ANSI SQL and MySQL	Overview of ANSI SQL, Introduction to MySQL, Importance of Standard SQL	https://www.dwbi.org/pages/30/introduction-to-ansi-sql
		https://www.w3schools.com/mysql/mysql_intro.asp
Data Retrieval with SELECT Statement	Basic SELECT syntax, Retrieving specific columns, Filtering data with WHERE clause, Sorting results with ORDER BY	https://www.geeksforgeeks.org/mysql-select-statement/
Filtering and Sorting Data	Using logical operators in WHERE clause, Combining conditions with AND, OR, Sorting data using multiple columns	https://www.sqlshack.com/learn-mysql-sorting-and-filtering-data-in-a-table/
Aggregate Functions and Grouping	Introduction to aggregate functions (SUM, AVG, COUNT, etc.), GROUP BY clause for grouping results, HAVING clause for filtering grouped data	https://www.geeksforgeeks.org/mysql-aggregate-function/
		https://www.w3schools.com/mysql/mysql_groupby.asp
Joins and Subqueries	INNER JOIN, LEFT JOIN, RIGHT JOIN, Self-joins and cross joins, Subqueries in SELECT, WHERE, and FROM clauses, Correlated Subqueries	https://www.geeksforgeeks.org/mysql-join-1/
		https://www.geeksforgeeks.org/mysql-subquery/
Data Modification with INSERT, UPDATE, DELETE	Inserting data into tables, Updating existing records, Deleting records from tables	https://www.geeksforgeeks.org/sql-ddl-dql-dml-dcl-tcl-commands/
Creating and Modifying Tables	Creating tables with CREATE TABLE, Modifying table structure with ALTER TABLE, Deleting tables with DROP TABLE	https://www.geeksforgeeks.org/sql-create-table/?ref=lbp
		https://www.geeksforgeeks.org/sql-alter-add-drop-modify/?ref=lbp
Indexes and Constraints	Creating indexes for better query performance, Defining primary and foreign key constraints, Ensuring data integrity with UNIQUE and CHECK constraints	https://www.w3schools.com/mysql/mysql_constraints.asp
		https://www.w3schools.com/mysql/mysql_create_index.asp

Check Your Understanding:

[MySQL Quiz](#)

Module 3

C# and ADO.NET

Overview:

This module offers an in-depth exploration of modern C# (up to version 12) and ADO.NET within the context of the .NET 8 ecosystem. Beginning with an overview of the .NET environment and hands-on setup guidance, learners will gradually build expertise in core programming concepts, object-oriented design, modern language features, multithreading, and secure coding practices. The module also dives into data access using ADO.NET, covering both connected and disconnected architectures. Practical labs and examples are designed to equip learners with real-world skills applicable in enterprise-level .NET applications.

Learning Objectives:

By the end of this module, learners will be able to:

- Explain the .NET platform, its versions, and install and configure development tools like .NET SDK and Visual Studio/VS Code.
- Develop and execute simple C# programs to understand the structure, syntax, and compilation process.
- Declare and use various C# data types, variables, constants, and apply type inference using `var` and `new()`.
- Build logical flows in programs using conditionals (`if`, `switch`) and loops (`for`, `while`, etc.), and leverage pattern matching and `switch` expressions.
- Create and invoke methods with various parameter types (`in`, `out`, `ref`), apply default values, and use method overloading and local functions.
- Design and implement classes, objects, constructors (including C# 12 primary constructors), and manage access control using modifiers.
- Use auto-implemented and `init-only` properties, define records for immutable data, and understand value equality.
- Apply inheritance, use `base`, override methods, and work with abstract classes and interfaces for polymorphic behavior.
- Manage runtime issues using `try-catch-finally`, custom exceptions, exception filters, null-handling operators, and C# 12's `required` modifier.

- Use core collections (List, Dictionary, etc.), iterate using foreach, and write LINQ queries for data filtering and transformation.
- Implement async programming using async/await, understand multi-threading concepts, handle concurrency issues, and use synchronization mechanisms.
- Use ADO.NET to connect to databases, retrieve, insert, update, and delete data using Connection, Command, DataReader, DataAdapter, DataSet, and DataTable.

Self-Learning (Open-source links):

C# and ADO.NET		
Topics	Sub-Topics	Learning Reference Links
Introduction to C# and .NET 8	Overview of .NET Ecosystem and Versions, Benefits of C# 12 Features in .NET 8, Setting up .NET SDK and Visual Studio Code/Visual Studio	https://learn.microsoft.com/en-us/dotnet/core/introduction
		https://learn.microsoft.com/en-us/dotnet/csharp/whats-new/csharp-12
		https://learn.microsoft.com/en-us/dotnet/core/install/windows
C# Syntax and Basic Constructs	Writing the First Program (Hello World), Variables, Constants, and Data Types (int, float, string, etc.), Value vs Reference Types, Type Inference with var and new()	https://www.tutorialspoint.com/csharp/csharp_basic_syntax.htm
		https://www.geeksforgeeks.org/c-sharp-data-types/
		https://www.tutorialsteacher.com/csharp/csharp-value-type-and-reference-type
		https://www.geeksforgeeks.org/c-sharp-variables/?ref=lbp
Control Flow Statements	Conditional Statements: if, else, switch (with enhancements in C# 12), Iterative Statements: for, while, do-while, foreach, Pattern Matching (switch expressions)	https://www.geeksforgeeks.org/c-sharp-decision-making-else-else-ladder-nested-switch-nested-switch/
		https://learn.microsoft.com/en-us/dotnet/csharp/fundamentals/functional/pattern-matching
Functions and Methods	Defining and Calling Methods, Parameters (in, out, ref) and Default Values, Method Overloading and Local Functions	https://learn.microsoft.com/en-us/dotnet/csharp/programming-guide/classes-and-structs/methods
		https://www.geeksforgeeks.org/c-sharp-method-overloading/

		https://www.geeksforgeeks.org/local-function-in-c-sharp/
Introduction to OOP Concepts	Classes and Objects, Constructors (Primary Constructors in C# 12) , Access Modifiers (public, private, protected, internal)	https://www.geeksforgeeks.org/introduction-of-object-oriented-programming/
		https://dev.to/ssukhpinder/primary-constructors-in-c-12-3ol1
Encapsulation and Properties	Auto-implemented Properties, init-only Setters and Records, Backing Fields	https://www.geeksforgeeks.org/c-sharp-encapsulation/
Inheritance and Polymorphism	Inheriting Classes and Using base Keyword, Overriding Methods and virtual/override Keywords, Abstract Classes and Interfaces	https://www.w3schools.com/cs/cs_polymorphism.php
		https://www.geeksforgeeks.org/difference-between-abstract-class-and-interface-in-c-sharp/
		https://www.w3schools.com/cs/cs_inheritance.php
Records and Structs	Value Equality and Immutability, Introduction to Records and With-expressions, Struct Improvements in C# 12	https://learn.microsoft.com/en-us/dotnet/csharp/fundamentals/object-oriented/
Exception Handling	try-catch-finally Blocks, Custom Exceptions and Exception Filters, when Clause in Exception Handling	https://www.geeksforgeeks.org/exception-handling-in-c-sharp/
Nullable Types and Null Safety	Handling Null References and ? Operator, Null-coalescing and Null-conditional Operators, Required Modifier in C# 12	https://www.geeksforgeeks.org/c-sharp-nullable-types/
		https://learn.microsoft.com/en-us/dotnet/csharp/language-reference/keywords/required
		https://www.geeksforgeeks.org/null-coalescing-operator-in-c-sharp/
Collections and LINQ	List, Dictionary, Queue, Stack Basics, Iterating Collections using foreach, LINQ Queries for Filtering and Transformation	https://learn.microsoft.com/en-us/dotnet/csharp/linq/get-started/introduction-to-linq-queries
		https://www.csharp.com/UploadFile/736bf5/collection-in-C-Sharp/

Tuples and Pattern Matching Enhancements	Creating and Deconstructing Tuples, Pattern Matching with is and switch Enhancements, List Patterns in C# 12	https://learn.microsoft.com/en-us/dotnet/csharp/fundamentals/functional/pattern-matching
		https://medium.com/codenx/c-12-collection-initializations-1e48009d2bc8
Asynchronous Programming with Async/Await	Introduction to Tasks and async/await, Exception Handling in Async Code, ValueTask vs Task Improvements in .NET 8	https://www.geeksforgeeks.org/async-and-await-in-c-sharp/
File Handling and I/O Operations	Reading and Writing Files, Working with Streams (FileStream, MemoryStream), JSON Serialization with System.Text.Json	https://www.geeksforgeeks.org/basics-of-file-handling-in-c-sharp/
Multi-Threading	What are Multi-Threading?, Importance of Multi-Threading, Multi-Threaded applications, System.Threading Namespace, System.Threading.Thread class, Process and Thread, C# Thread Life Cycle, Obtaining Current Thread Information, Thread Creation, Background Thread, Concurrency issues - Race Condition, Deadlocks, Synchronization, Locks, Monitor, Using [Synchronization] Attribute, Mutex, TaskScheduler Class	https://www.geeksforgeeks.org/c-sharp-multithreading/
		https://essentialcsharp.com/task-schedulers-and-the-synchronization-context
Debugging and Tracing	What is Debugging?, Why do I need to debug?, What is Tracing?, Debug/Trace Class Members, Assert Overloads, Configuration file EXE.CONFIG , Conditional Debug Attributes, Logging with System.Diagnostics.Trace, Logging with log4net	https://learn.microsoft.com/en-us/troubleshoot/developer/visualstudio/csharp/language-compilers/trace-and-debug
		https://stackify.com/log4net-guide-dotnet-logging/
Secure Coding	What's Secure Code?, XML vulnerabilities, XSS, basics of web security, authentication and authorization in ASP.NET, SQL Injection, Broken Authentication and Session Management,	https://learn.microsoft.com/en-us/dotnet/standard/security/secure-coding-guidelines

	Cross-site scripting, Security Misconfiguration, Cross-site request forgery (CSRF)	
ADO.NET Fundamentals	What is ADO.NET?, Why we need ADO.NET?, Various Connection Architectures, ADO.NET and it's class library, Connection Class, Command Class, DataReader Class, DataAdaptor Class, DataSet Class, DataTable Class, How to Connect to a Database using ADO.NET?, How to retrieve and modify data in database using ADO.NET?	https://www.geeksforgeeks.org/introduction-to-ado-net/

Check Your Understanding:

[C# Quiz](#)

[ADO.NET Quiz](#)

What Next

Congratulations on successfully completing the DN 5.0 Upskilling Learning Program!

As you have now finished this important phase of your learning, you will be taking a **Skill Based Assessment (SBA)** to certify your skills. **This assessment will cover all the skills and topics you have learned during Upskilling Learning Program**, ensuring you have a comprehensive understanding of the material.

We wish you the best of luck for your assessment and look forward to seeing you apply your newly acquired knowledge and skills. Good luck!